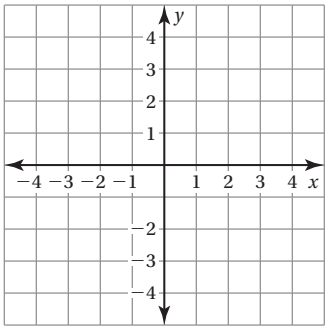
Name _____ Date _____

Standards-Based Practice

Geometry:
Polygons in the Coordinate Plane

Draw polygons in the coordinate plane given coordinates for the vertices; use coordinates to find the length of a side joining points with the same first coordinate or the same second coordinate. Apply these techniques in the context of solving real-world and mathematical problems.

1. Draw the polygon with vertices $A(-3, 3)$, $B(-3, 2)$, and $C(1, 1)$ in the coordinate plane.

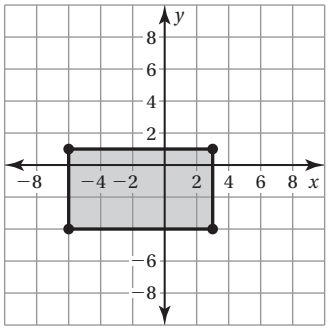


What is the length of $\overline{AB}$?

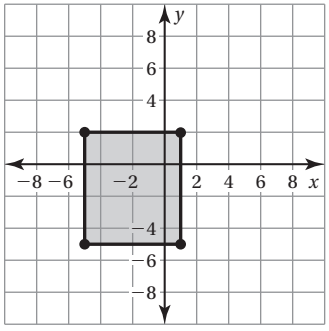
---

2. Which rectangle has the greatest area?

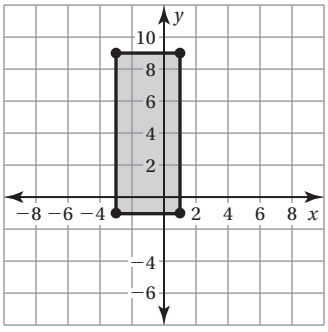
A.



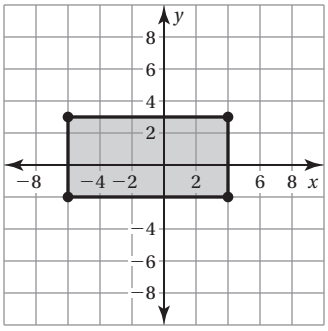
B.



C.



D.



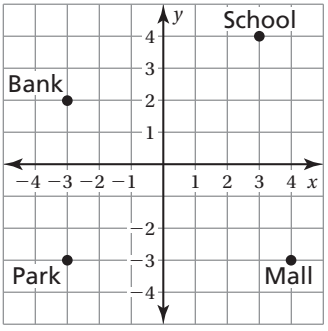
Geometry

Copyright © Big Ideas Learning, LLC.
All rights reserved.

3. Find the distance between point $A(-7, -11)$ and point $B(4, -11)$.

---

4. The grid represents the sidewalks in part of a city. Each unit square represents one city block. It takes 4 minutes to walk the length of one city block.



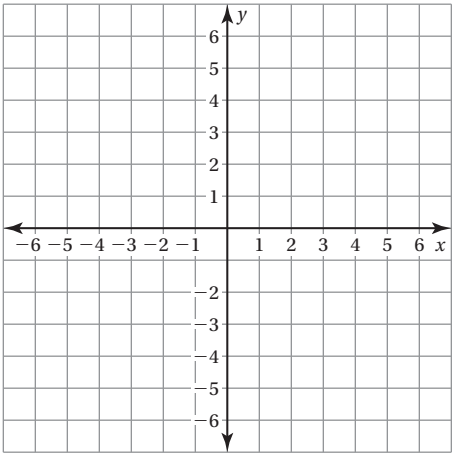
a. What is the least amount of time it takes to walk from the mall to the bank using only the sidewalks?

b. You walk from the mall, to the park, to the bank. Your friend walks from the mall, to the school, to the bank. Who takes longer to reach the bank? How many minutes longer?

c. The location of a grocery store was chosen so that it forms a rectangle with the mall, park, and bank. What is the location of the grocery store?

---

5. Draw a polygon with vertices $A(-3, -4)$, $B(-3, 3)$, $C(0, 6)$, $D(3, 3)$, and $E(3, -4)$ in the coordinate plane. Label point $F(0, -4)$. Connect point B and point D.



a. What is the distance between point D and point E?

A. 3 units **C.** 7 units

B. 4 units **D.** 8 units

b. What is the distance from point C to point F?

---

6. Find the distance between point $M(4, 8)$ and point $N(4, -6)$.

Geometry

Copyright © Big Ideas Learning, LLC.
All rights reserved.